

Algebra B - T1 Minimal Polynomial and Diagonalization

1. Basic Properties of the Minimal Polynomial

Let $A \in M_n(F)$, and let m_A denote the minimal polynomial of A .

Proposition 1.1 (Divisibility Property)

The minimal polynomial m_A divides every polynomial that annihilates A .

In particular,

$$m_A \mid \chi_A$$

where χ_A is the characteristic polynomial of A .

Proposition 1.2 (Roots)

The polynomials m_A and χ_A have exactly the same roots, possibly with different multiplicities.

Proposition 1.3 (Field Extension)

The minimal polynomial m_A does not change if we enlarge the underlying field.

2. Diagonalizability Criterion

Proposition 2.1

A matrix $A \in M_n(\mathbb{C})$ is diagonalizable if and only if its minimal polynomial has only simple roots.

Equivalently,

$$A \text{ is diagonalizable} \iff m_A \text{ splits into distinct linear factors.}$$

3. Example: A Matrix in $M_3(\mathbb{R})$

Problem

Let $A \in M_3(\mathbb{R})$ satisfy

$$\det(A) = -2 \quad \text{and} \quad A^4 + 2A^3 + A^2 - 2A - 2I = 0.$$

Determine whether A is diagonalizable over $\mathbb{R}$.

Step 1: Annihilating Polynomial

Define

$$P(X) = X^4 + 2X^3 + X^2 - 2X - 2.$$

Then $P(A) = 0$, so P annihilates A . Hence,

$$m_A \mid P.$$

Step 2: Factorization of P

We compute:

$$P(1) = 0, \quad P(-1) = 0,$$

so $X - 1$ and $X + 1$ are factors.

Thus,

$$P(X) = (X - 1)(X + 1)(X^2 + 2X + 2).$$

Observe that

$$X^2 + 2X + 2$$

has no real roots (its discriminant is negative), and is therefore irreducible over $\mathbb{R}$.

Step 3: Consequences for m_A

Since $m_A \mid P$, the roots of m_A must be among

$$1, -1 \cup \text{complex roots of } X^2 + 2X + 2.$$

If A were diagonalizable over $\mathbb{R}$, then m_A would split into distinct linear factors over $\mathbb{R}$. Hence its roots must be real, so they must be contained in

$$1, -1.$$

Step 4: Using the Determinant

Recall that for $A \in M_3(\mathbb{R})$,

$$\det(A) = \lambda_1 \lambda_2 \lambda_3,$$

where λ_i are the eigenvalues.

If $m_A = X - 1$, then

$$\chi_A = (X - 1)^3,$$

which implies

$$\det(A) = 1,$$

contradicting $\det(A) = -2$.

Thus,

$$m_A \neq X - 1.$$

A similar argument excludes the possibility that all eigenvalues are ± 1 .

Step 5: Conclusion

The minimal polynomial must involve the irreducible factor

$$X^2 + 2X + 2,$$

which has no real roots.

Therefore, m_A does not split over $\mathbb{R}$, and hence:

Final Conclusion

A is not diagonalizable over $\mathbb{R}$.

4. Relationship Between f and f^2

Let $f \in \mathcal{L}(V)$.

Proposition 4.1

If f is diagonalizable, then f^2 is diagonalizable.

Proof.

If $[f]_{\mathcal{B}} = D$ is diagonal, then

$$[f^2]_{\mathcal{B}} = D^2,$$

which is also diagonal.

Proposition 4.2 (Converse Direction)

Suppose f^2 is diagonalizable.

Then the minimal polynomial m_{f^2} has only simple roots:

$$m_{f^2}(X) = \prod_{i=1}^r (X - \lambda_i^2).$$

Define

$$P(X) = \prod_{i=1}^r (X^2 - \lambda_i^2).$$

Then $P(f) = 0$, so

$$m_f \mid P.$$

Now factor:

$$X^2 - \lambda_i^2 = (X - \mu_i)(X - \overline{\mu_i}),$$

so P splits into distinct linear factors over $\mathbb{C}$.

Hence m_f also has simple roots, and therefore:

Conclusion

f is diagonalizable over $\mathbb{C}$.

Remark

If f is invertible, then all eigenvalues $\lambda_i \neq 0$, so the above argument is valid without degeneracy.

5. Final Observations

- $\deg(\chi_A) = 3$, so χ_A must have at least one real root
- Since m_A and χ_A share the same roots, m_A must also have at least one real root
- Therefore, the polynomial $X^2 + 2X + 2$ alone cannot be the minimal polynomial